

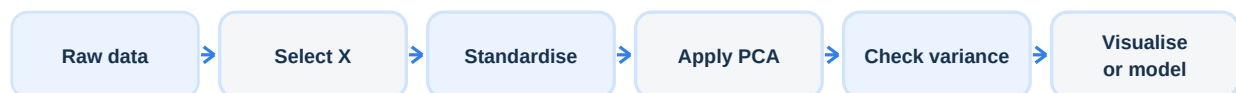
PCA Theory Quick Reference

Dimensionality Reduction with PCA - structured concepts, simple explanations, and notebook-ready memory lines

PCA in one sentence: PCA finds the best new summary directions so we can describe many features with fewer components while keeping as much important spread as possible.

How to use this PDF

Use this as a quick reference before coding. Read the concept name, the simple meaning, and the memory line. The goal is clarity first, mathematics second.



Day 5 task map

Concept	Simple meaning	Tiny example / memory line
Beginner	Wine Quality PCA for 2D and 3D visualisation.	Standardise -> PCA(2) -> scatter -> variance bar -> PCA(3).
Intermediate	RNA-Seq PCA for optimal components and faster modelling.	All components -> scree plot -> 95% variance -> Logistic Regression.
Expert	Compare PCA, t-SNE, UMAP, runtime and trustworthiness.	PCA vs t-SNE vs UMAP, then PCA(50) + UMAP.

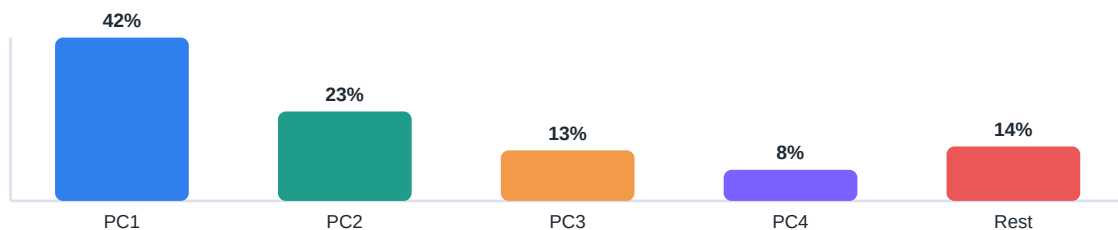
1. Foundation Concepts

These words make PCA understandable. If these are clear, the rest becomes much easier.

Concept	Simple meaning	Tiny example / memory line
Dimension	One feature or column in the dataset.	Wine has alcohol, pH, density, sulphates. Each is one dimension.
High-dimensional data	Data with many columns/features.	RNA-Seq can have thousands of gene columns.
Curse of dimensionality	Too many features make data harder to visualise, slower to train, and sometimes noisier.	More columns can mean more confusion, not always more insight.
Direction	The way the data mainly moves or changes.	Study hours go up and exam score also goes up: that is an upward direction.
Spread	How far apart values are.	Heights 170,171,172 have small spread; 150,170,195 have large spread.
Noise	Random messy variation that hides the main pattern.	A student studies a lot but scores lower due to bad sleep. That messy effect is noise.
Variance	The mathematical word for spread.	High variance means values are widely spread out.
Standardisation	Putting all features on a fair scale before PCA.	Mean becomes 0 and standard deviation becomes 1.

**Memory: Direction = which way? Spread = how much? Noise = messy extra movement.
Variance = measured spread.**

Example: PCA keeps the largest spread first



2. What PCA Creates

PCA does not simply delete original columns. It creates new summary columns called principal components.

Concept	Simple meaning	Tiny example / memory line
Principal component	A new PCA-created feature made from a mixture of original features.	PC1 is not alcohol or pH; it is a new summary column.
PC1	The first principal component. It captures the largest spread.	The strongest new direction.
PC2	The second principal component. It captures the next largest spread and is at a right angle to PC1.	The second-best direction.
PC3 and beyond	More components that capture smaller and smaller amounts of spread.	Useful for modelling when 2D is not enough.
Linear combination	A weighted mixture of original features.	$PC1 = 0.45 \cdot \text{alcohol} - 0.40 \cdot \text{density} + \dots$
Loading	The weight of an original feature inside a component.	A high alcohol loading means alcohol strongly helps build that PC.
Score	The new PCA coordinate of one row/sample.	Wine A may become $PC1 = 1.84$, $PC2 = -0.72$.
Projection	Moving each original data row onto the PCA axes.	Many wine columns become a new 2D address: PC1 and PC2.

Memory: Loadings explain how features build the axes. Scores explain where rows land on those axes.

Mini example

Original wine row: alcohol, pH, density, sulphates, chlorides, sugar, acidity, ...

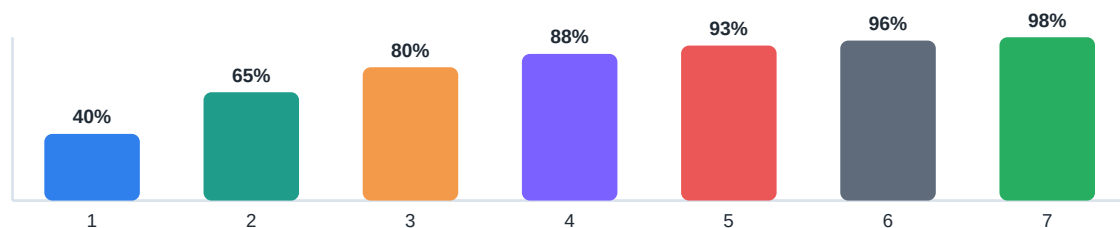
After PCA projection: PC1 score, PC2 score. These two values can be plotted as a 2D scatter plot.

3. Reading PCA Outputs

After PCA, we need to know whether the reduced version kept enough information.

Concept	Simple meaning	Tiny example / memory line
Explained variance	How much spread a component captures.	PC1 captures the biggest share.
Explained variance ratio	The percentage/share of total spread captured by one PC.	0.42 means 42% of the total spread.
PC1 + PC2 total	How much information is kept in a 2D PCA plot.	If PC1 = 32% and PC2 = 21%, the 2D plot keeps 53%.
Cumulative variance	The running total after adding components.	PC1 + PC2 + PC3 = total kept by first 3 PCs.
Scree plot	A chart showing explained or cumulative variance across components.	Used to decide how many PCs to keep.
Elbow	The point where adding more components gives much smaller benefit.	Fast rise first, then flattening.
95% rule	Choose the smallest number of components that keeps at least 95% variance.	Common for modelling compression, not for simple 2D plots.

Example: cumulative explained variance



Memory: Explained variance ratio = information in one PC. Cumulative variance = information after adding PCs together.

4. PCA for Visualisation vs Modelling

PCA has two common goals. The number of components depends on the goal.

Concept	Simple meaning	Tiny example / memory line
Visualisation PCA	Use PCA to make a human-readable picture.	Usually keep 2 PCs for 2D or 3 PCs for 3D.
2D PCA scatter	Plot PC1 on x-axis and PC2 on y-axis.	Colour by wine quality or tumour type.
3D PCA scatter	Plot PC1, PC2, and PC3.	Keeps a bit more structure than 2D.
Modelling PCA	Use PCA to reduce many columns before training a model.	Often keep enough PCs for 90-95% variance.
Faster training	Fewer columns usually means less computation.	RNA-Seq: thousands of genes -> fewer PCA components.
Accuracy check	PCA may or may not improve prediction.	Always compare model accuracy and runtime.

For visualisation, 2 components can be useful even if they do not keep 95%. For modelling, we usually keep many more components.

Notebook interpretation template

A PCA plot gives a simplified view of the feature space. If classes overlap, it does not mean PCA failed; it means the first two variance directions do not fully separate the labels.

5. Interpretation and Limitations

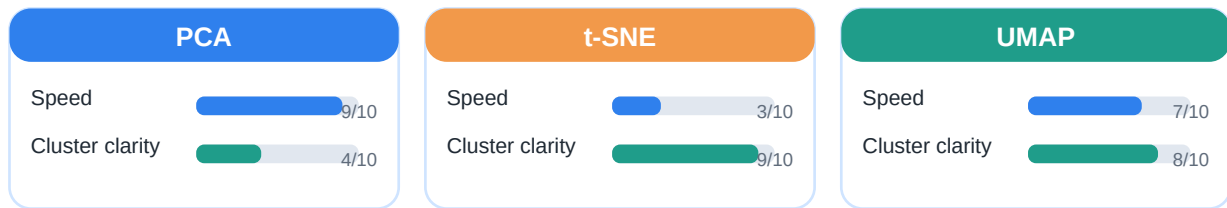
PCA is powerful, but it must be interpreted carefully.

Concept	Simple meaning	Tiny example / memory line
Positive loading	The feature pushes the component in one direction.	Positive does not mean good.
Negative loading	The feature pushes the component in the opposite direction.	Negative does not mean bad.
Absolute loading	The strength of a feature's contribution, ignoring sign.	Large absolute value = strong contribution.
Unsupervised	PCA does not use the target label when creating PCs.	Wine quality is used only for colouring after PCA.
Linear method	PCA finds straight directions.	Curved patterns may be better shown by UMAP/t-SNE.
Information loss	Dropped components may contain detail.	PC1 + PC2 = 60% means 40% is outside the 2D view.
Scale sensitive	Big raw-number columns can dominate if not standardised.	Always standardise numeric features first.
Outlier sensitive	Extreme points can pull PCA directions.	Check data quality before PCA.
Variance vs prediction	High variance does not always mean best for prediction.	PCA must be tested with model performance.

Professional wording: This feature contributes strongly to PC1. Avoid saying: This feature is the most predictive, unless a supervised model proves it.

6. PCA vs t-SNE vs UMAP

All three reduce dimensions, but they preserve different things and have different trade-offs.

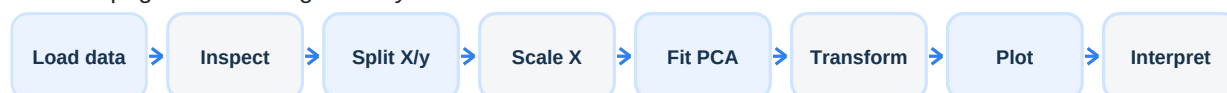


Concept	Simple meaning	Tiny example / memory line
PCA	Linear, fast, keeps maximum variance.	Best straight view of the data.
t-SNE	Non-linear, focuses strongly on keeping close neighbours close.	Often shows beautiful clusters, but can be slow.
UMAP	Non-linear, keeps neighbourhood structure and is often faster than t-SNE.	Popular for high-dimensional biological data.
Local structure	Nearby-neighbour relationships.	Who is close to whom?
Global structure	The broad overall arrangement of the dataset.	Big map-level trends.
Perplexity	t-SNE neighbourhood-size setting.	Perplexity = 30 is a common starting point.
n_neighbors	UMAP neighbourhood-size setting.	Small value = more local focus.
Trustworthiness	Checks whether 2D neighbours were truly neighbours in original space.	Higher score = more reliable local neighbourhoods.
PCA + UMAP	Compress with PCA first, then apply UMAP.	Thousands of genes -> 50 PCs -> 2D UMAP.

Memory: PCA = best straight view. t-SNE = neighbour clusters. UMAP = neighbour clusters with practical speed. PCA + UMAP = compress first, then map.

7. Notebook Workflow Checklist

Use this page while coding the Day 5 PCA notebook.



Concept	Simple meaning	Tiny example / memory line
1. Load data	Read the CSV and inspect shape, columns, missing values.	<code>df.shape, df.head(), df.info()</code> .
2. Select X and y	Use numeric input features as X and label as y.	Wine: X = chemicals, y = quality.
3. Standardise X	Fit StandardScaler and transform the features.	PCA should not be fooled by scale.
4. Fit PCA	Choose n_components based on goal.	2 for 2D, 3 for 3D, or enough for 95%.
5. Transform data	Create PCA scores for every row.	<code>pca.fit_transform(X_scaled)</code> .
6. Plot scores	Scatter PC1 vs PC2, colour by class.	Look for clusters, overlap, and outliers.
7. Check variance	Use explained_variance_ratio_ and cumulative variance.	Report how much information was kept.
8. Interpret loadings	Find which features or genes build PC1/PC2 most strongly.	Use absolute loadings for contribution strength.
9. Model comparison	For modelling, compare accuracy and runtime before/after PCA.	Do not assume PCA improves accuracy.

Final mini summary

PCA is a smart compression method: it rotates the feature space, finds the strongest directions of spread, projects each row onto those directions, and lets us visualise or model with fewer dimensions.